Smart AI-Based Traffic Management System

Using Computer Vision

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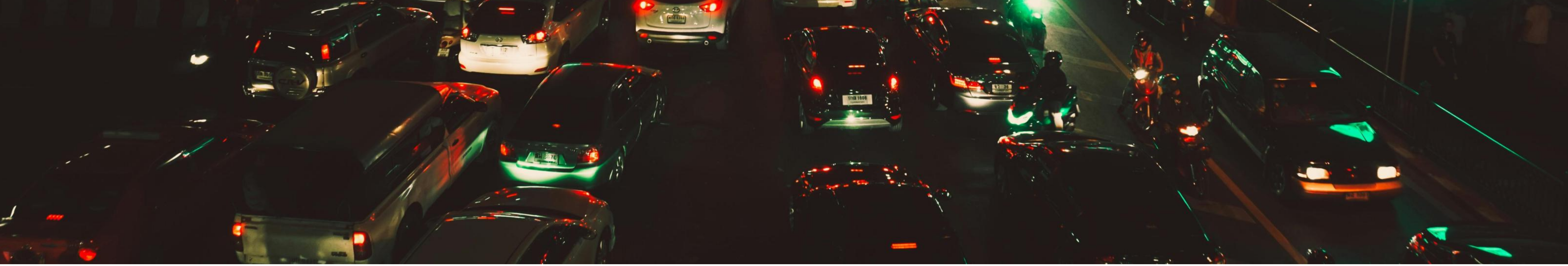
Problem Statement:

Smart Traffic Management System for Urban Congestion

Design an AI-based traffic management system to optimize signal timings and reduce congestion in urban areas. The system should analyze real-time traffic data from cameras and IoT sensors to predict and mitigate bottlenecks.

Economic Drain

Long waiting times and inefficient traffic movement result in lost productivity and economic wastage for commuters and businesses alike.



The Challenge: Dynamic Traffic Demands

1

Multi-Directional Bottlenecks

Complex intersections often experience congestion from multiple directions simultaneously, overwhelming static signal timings.

2

Emergency Vehicle Delays

Current systems struggle to provide immediate priority to emergency vehicles, compromising critical response times.

3

Traditional System Limitations

Outdated infrastructure lacks the intelligence to adapt to real-time changes, leading to persistent inefficiency and frustration.

Key Objectives for Smarter Mobility

Optimise Waiting Times

Significantly reduce vehicle idle time at intersections through intelligent signal control.

Improve Traffic Throughput

Enhance the overall flow of vehicles, maximising road capacity and efficiency.

Minimise Emissions

Decrease fuel consumption and harmful emissions by reducing stop-and-go traffic.

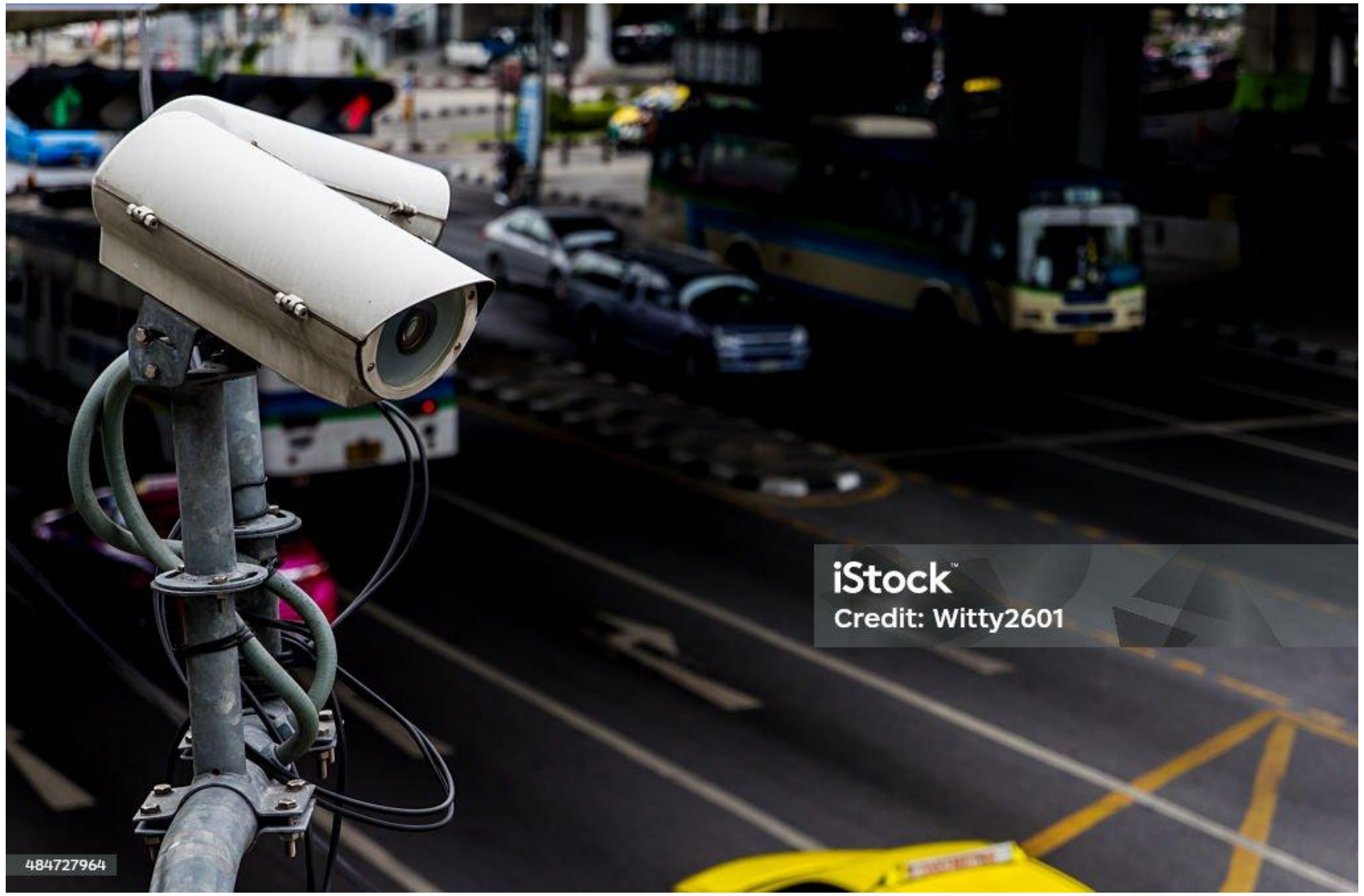
Prioritise Emergency Services

Ensure swift passage for emergency vehicles, cutting down response times.

Enhance Road Utilisation

Achieve more equitable and efficient use of existing road networks.

Essential Data for Intelligent Traffic Flow



CCTV Camera Feeds

Real-time video streams are crucial for dynamic analysis and monitoring traffic conditions.

Vehicle Type Classification

Differentiating cars, buses, and trucks helps in prioritising and optimising signal timings.

Lane-Wise Density

Understanding congestion levels in individual lanes enables precise signal adjustments.

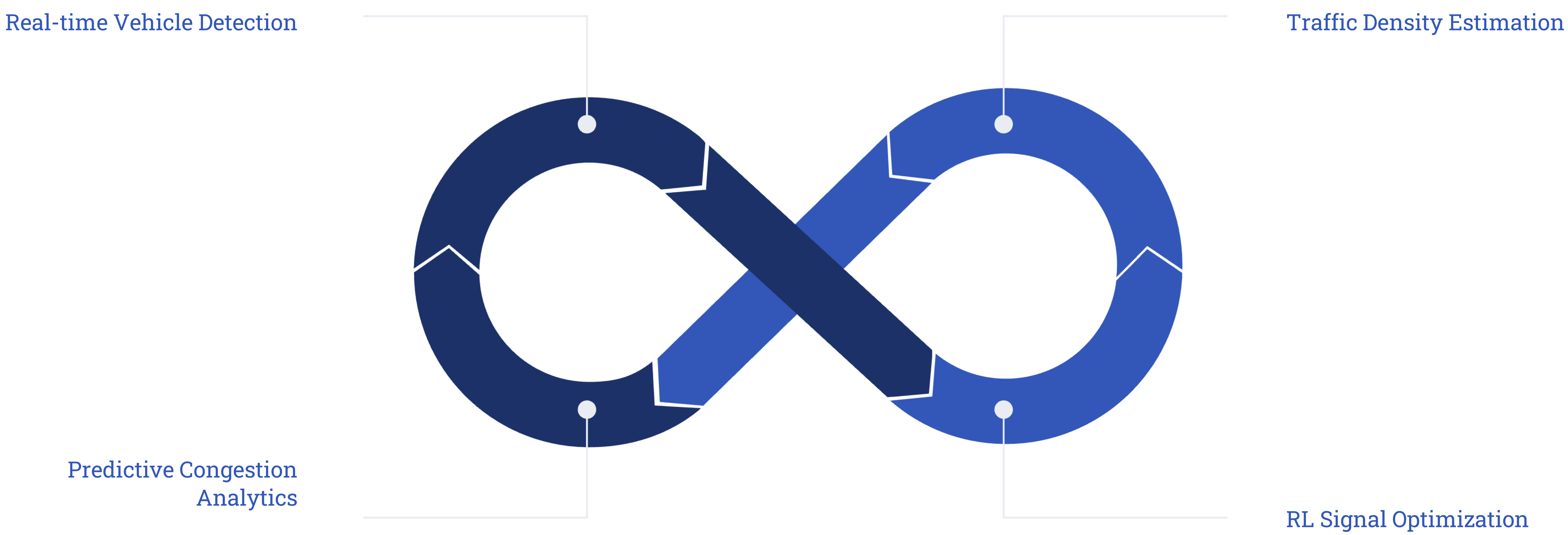
Historical Traffic Patterns

Leveraging past data helps predict peak hours and recurring congestion points.

Weather & Event Data

External factors like weather or special events significantly impact traffic, requiring adaptive responses.

Our Proposed AI-Driven Solution



Real-time Vehicle Detection

Utilising YOLO/CNN models for accurate and instant identification of vehicles on the road.

Traffic Density Estimation

Precise calculation of vehicle concentration to inform signal adjustments.

RL-based Signal Optimisation

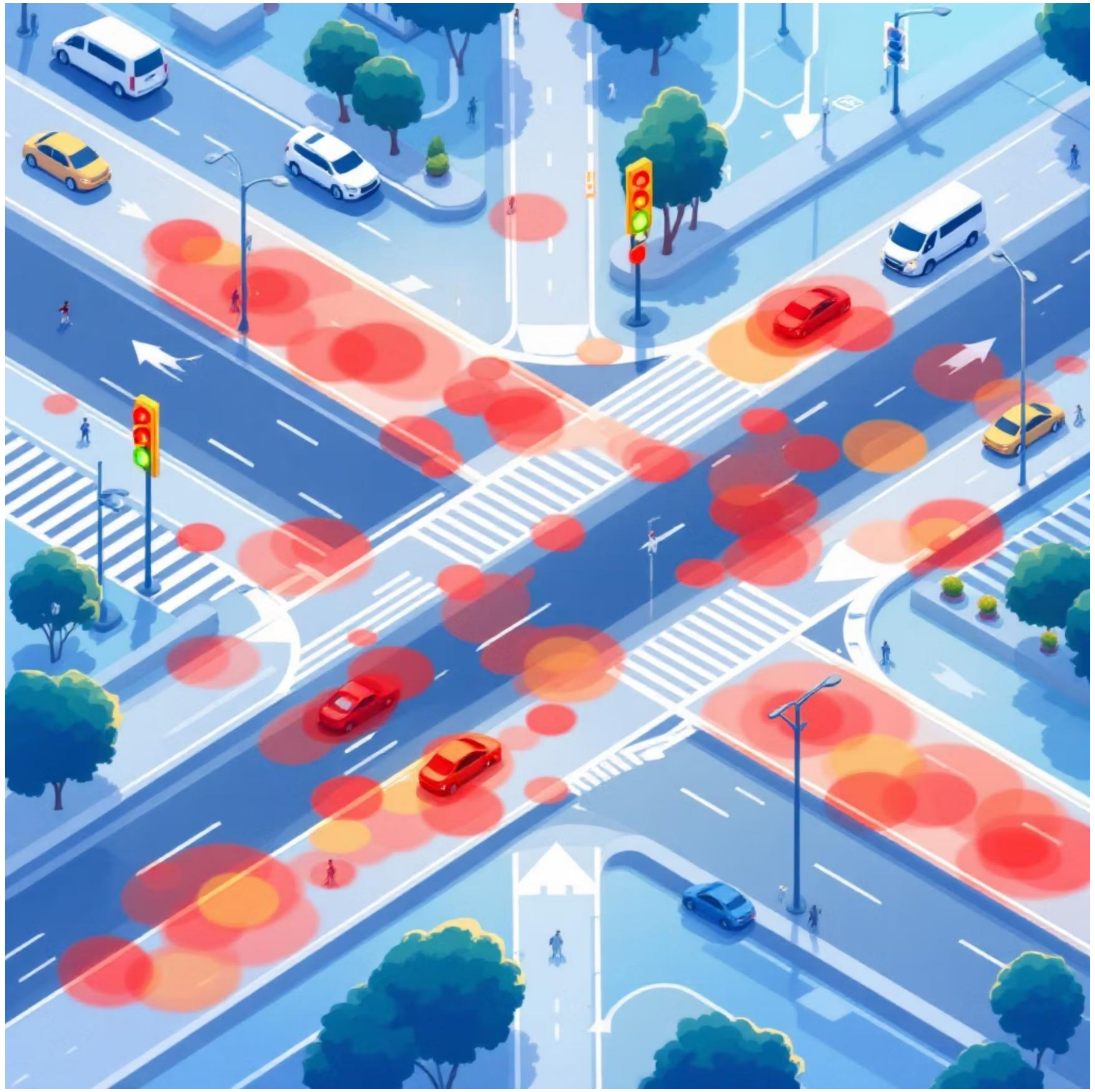
Reinforcement Learning algorithms dynamically adjust signal timings for optimal flow.

Predictive Congestion Analytics

Anticipating future traffic build-ups to proactively manage flow and prevent gridlock.

Core System Functionalities

1	**Vehicle Counting** Accurate enumeration of vehicles for real-time traffic volume assessment.
2	**Lane-Wise Density Heatmaps** Visual representation of traffic congestion across individual lanes for granular analysis.
3	**Dynamic Green-Time Allocation** Adaptive adjustment of green light durations based on current traffic demand.



AI & Analytics: The Engine of Efficiency

YOLOv8 & Vision Models

State-of-the-art computer vision models for robust object detection and classification.

Reinforcement Learning (DQN/PPO)

Algorithms that learn optimal signal timing strategies by continuous interaction with the traffic environment.

Data Sources

Use open traffic video datasets
UA-DETRAC ,DATS_2022 (Indian roads) ,IITM-HeTra

Historical Trend Analysis

Continuous learning from past data to refine predictive models and adapt to seasonal or daily patterns.

AI & Analytics: The Engine of Efficiency

•**Collect Data**
•Use public traffic video datasets + small locally recorded videos.
•Generate simulated traffic data using SUMO.

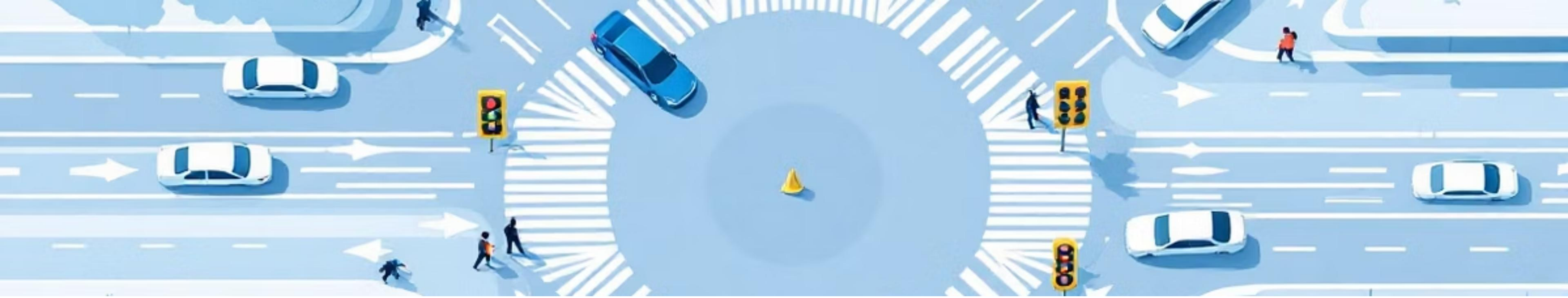
•**Process Video (Computer Vision)**
•Use YOLO to detect vehicles in each frame.
•Count vehicles and estimate traffic density per lane.

•**Create Traffic State**
•Build a state vector with vehicle count, queue length, waiting time, and current signal phase.

•**AI Decision System**
•Use ML/Reinforcement Learning to predict the optimal green-light duration.
•Choose the direction with the highest traffic density.

•**Adaptive Signal Control Loop**
•Continuously update signal timings based on new video input.
•Recalculate every few seconds.

•**Evaluation**
•Test in SUMO simulator.
•Compare AI-based dynamic control vs fixed-timer control.
•Measure waiting time, queue length, and fuel/CO_2 reduction.



Multi-Directional Traffic Handling

1

Preventing Bottlenecks

Proactive management of intersections ensures no single direction dominates, preventing gridlock.

2

Smooth Phase Transitions

Seamless shifts between green light phases, minimising sudden stops and accelerations.

3

Intelligent Lane Priority

Allocating green time based on real-time lane occupancy and vehicle queues.

Expected Outcome: A Prototype in Action

The project will deliver a fully functional software prototype demonstrating the system's capabilities.

Simulated Urban Environment

Testing and validation in a realistic virtual city.

Traffic Management Dashboard

Centralised control and monitoring for operators.

Proven Efficiency

Evidence of 10% commute time reduction.

Future Scope: Expanding Impact

Beyond the prototype, our vision extends to broader applications and enhanced functionalities.

Adaptive Public Transport

Integrating with bus and tram schedules for holistic optimisation.

Emergency Vehicle Prioritisation

Ensuring clear pathways for critical services.

Pedestrian & Cyclist Safety

Incorporating vulnerable road user considerations.

Environmental Impact Tracking

Monitoring emissions reduction due to smoother traffic.

Conclusion: A Smarter City Awaits

Our AI-powered traffic management system offers a tangible solution to urban congestion, paving the way for more efficient, sustainable, and liveable cities.

Thank You